

Case study

Efficiency of convolutional neural networks (CNN) based image classification for monitoring construction related activities: A case study on aggregate mining for concrete production

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ABSTRACT

Monitoring construction activities is an important task for efficiency in construction site operations thus the topic received a fair amount of attention in the literature. Optimizing construction site operations by monitoring and detecting various tasks is dependent on the size of the construction field, which determines the tools that can be used for the job. A monitoring task can be performed with high efficiency through image classification algorithms by training the algorithms to detect construction machinery. If the area of monitoring is larger, such as the task of detecting construction related operations in a large infrastructural construction, using drone images might become inefficient. We aimed to take a first step towards a cost-efficient monitoring system specifically for construction activities that cover large territories. Consequently, satellite image classification has been performed for construction machinery detection in this study. We utilized different versions of well-established convolutional neural network architectures as backbone for the transfer learning method and their performances are evaluated. Finally, the best performing models are determined as DenseNet161 and ResNet101 with 0.919 and 0.903 test accuracies, respectively. DenseNet161 model was discussed in terms of its accuracy and efficiency in a case study to detect illegal aggregate mining activity through the basin of Thamarabarani River.

1. Introduction

Developing civil engineering applications with artificial neural networks become important in many different areas in the literature. Some of these studies include predicting diameters of jet grouted columns [1], prediction of the residual flexural strength of fiber reinforced concrete [2], and predicting drying shrinkage of alkali-activated blast furnace-fly ash mortars [3]. A specific application of artificial neural networks is image processing and classification which also have a potentially broad application area in civil engineering problems. The continuous development of computer vision and deep learning algorithms over the years provides researchers a broad perspective for new fields of study.

One of the most important developments in computer vision studies is transfer learning. Transfer learning enables researchers to utilize neural networks previously trained on very large datasets to be used in completely different detection or classification jobs with much smaller datasets. The method became highly popular since it reduces training time drastically [4]. General features of images are learned in the previous layers by training the neural network using a large dataset such as Coco [5], ImageNet [6], etc. The trained

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weights of these lower layers are directly used as non-trainable weights and the last classification layer is replaced with a freshly initialized classification layer. This new classification layer is trained using a smaller dataset. Since the model can already detect main features, only the final classification layer is trained which increases efficiency significantly. ImageNet is one of the most popular datasets that is commonly used for transfer learning containing more than a million images. ImageNet's popularity comes from the fact that it is one of the first large-scale datasets to pioneer image classification studies and it is proved to be efficient, especially for transfer learning [7,8]. The development of computer vision studies trained using the ImageNet dataset has started with SIFT + FVs [9]. This was accelerated with AlexNet [8] and continued with VGG-19 [10], Inception V2 [11], ResNeXt-101 [12], and so on.

1.1. Related work

Civil engineering applications of image classification and object detection mostly involve images obtained either from Unmanned Aerial Vehicles (UAVs) or satellites. UAV images are commonly used in various industries such as mining, agriculture, and aerial mapping as well as in the construction industry for the analysis of structural elements, safety and quality monitoring, and maintenance control [13]. A well-known application of image processing is on-site safety monitoring against accidents involving construction machinery [14]. Detection of construction machinery using image classification also has the potential to be used in a range of monitoring practices ranging from construction activity monitoring to detecting illegal construction, illegal mining, and unlicensed logging in forests. Transfer learning is used for most of the construction machinery detection literature based on UAV images [15–17]. The use of transfer learning in image classification studies is popular because it is both easy and fast to work with pre-trained models [18].

Construction activity monitoring for construction sites is an important task for efficiency and cost of operations. There are a few studies utilizing transfer learning to detect construction machinery from UAV images as an initial step for construction activity monitoring. Arabi et al. proposed the SSD_MobileNet object detection model to detect construction vehicles. They used large-scale datasets such as ImageNet, Common Objects in Context (COCO), and Open Image. They also used the AIM dataset with construction machine images. This dataset, which is a subset of ImageNet, contains the street view of excavators, loaders, rollers, concrete mixer trucks, and dump trucks images. The mean average precision (mAP) value is 91.20% [19].

Another study conducted by Fang et al. [20] aimed to detect workers and excavators on construction sites using IFaster R-CNN (Improved Faster Regions with Convolutional Neural Network Features). They created a dataset with UAV images collected from different construction sites to train the Convolutional Neural Network (CNN) model. The accuracy of the proposed model for excavators was 95 %. Precision and Recall values were 99% and 81%. Guo et al. proposed cameras placed at an angle that capture the construction site environment can make environmental risk analysis and classification of construction vehicles possible. The average precision value obtained was 98.8 % [21].

As mentioned earlier satellite images are far more efficient for covering large areas, they provide low-cost access to aerial imagery over large areas without the need for the on-site presence of manpower as opposed to UAV applications [22]. Images collected from satellites have a wider range of earth imaging capabilities compared to UAVs and they are also commonly used by multiple disciplines as well as civil engineering [23,24]. Consequently, performing satellite image processing with convolutional neural networks has gained popularity among researchers. Image classification with satellite images enables remote sensing data processing so it has become a broad research area [25]. Classification algorithms are commonly utilized with satellite imagery for land classification [26], image scene classification [27], vehicle detection [21], and tree species classification [28].

Among the classification algorithms, CNNs are the most popular group in various domains as well as civil and environmental engineering. A recent example for the use of satellite images for remote sensing image classification with convolutional neural networks would be Sharma et al. [29]. The land cover classification was performed with Landsat 8 satellite images. The deep patch-based CNN system they proposed achieved 85.60 % accuracy. Pereire et al. [4] performed large-scale active fire detection using Landsat-8 satellite images. In Pereire et al.'s study. They obtained 87.2 % precision and 92.4 % recall. Another study worth mentioning is Nogueira et al. [15] which compared various architectures for remote sensing scene classification using satellite images. Remote sensing scene classification aims to label images collected with earth observation technology according to their semantic categories [27]. AlexNet and VGG16 yielded the highest average accuracy values of over 93 % compared to global descriptors.

1.2. Contribution

Satellite image classification is mostly used for land or image scene classification simply because large areas can only effectively be monitored with satellite imagery. As mentioned earlier few studies that utilized transfer learning for construction vehicle detection exclusively used UAV images [19–21]. However current ranges of UAVs restrict their use when large territories need to be monitored. To the best of the authors' knowledge, no study classifies satellite images to detect construction machinery although few studies in the literature perform construction machinery classification using drone images. This study covers classification of satellite images based on existence of construction machinery. We aim to build the foundation of future methodologies designed to monitor large areas for construction related activities. Using satellite images could enable monitoring construction and construction related activities for very large areal zones specifically infrastructural construction activities such as dam construction. It can also be potentially used for detecting illegal construction and unlicensed logging activities since this requires very large territories to be scanned.

Illegal mining is a widespread problem with a direct impact on global environmental problems [30]. The construction industry is mostly responsible for illegal aggregate mining, constituting a large portion of illegal mining, which is a vital environmental problem for many countries. Illegal aggregate mining causes water pollution by destroying drinking water sources and deforestation by

damaging the ecological environment [30]. As the urban population increases, the need for larger infrastructures also increases. Sand extracted from the lake and riverbeds is used as a quality building material in concrete production [31]. More specifically aggregate is used in large quantities in the construction industry specifically for concrete production, facilitating a high demand for a cheap alternative source of aggregate. A large portion of this demand is compensated by illegal aggregate mining which destroys agricultural lands, threatening the economy of developing countries. There is no solid data on the amount of aggregate obtained by illegal aggregate mining because there is no regulated legal process in this industry [31,32]. To prevent these problems, legal regulations have been initiated in countries such as China and India. Although the existence and consequences of illegal aggregate mining are known, these operations are still difficult to detect within the framework of legal rules [30,31] because they are carried out irregularly and current regulations are hard to apply in practice.

Since illegal aggregate mining activities are difficult to detect, using machine learning applications for construction machinery detection from satellite images can be a solution in regions for high-risk illegal aggregate mining activity areas. Satellite images taken at different times are known to be effective for change detection [33]. Accordingly, this study proposes a neural network model trained using a dataset of Google Earth images that can detect images containing construction machinery with high accuracy. We used image classification for the task of detecting construction machinery because it is not necessary to detect the number or coordinates of the machinery. It is enough to flag an image as positive if it contains any construction machinery for the tasks such as change detection for high-risk areas of illegal aggregate mining. Additionally, the largest area covered in one image in the dataset is around 0.19 km² (300 m camera object distance). Once an image is flagged as positive, the location of the machinery can easily be determined using the georeferencing information in file formats commonly used for satellite images such as GeoTIFF.

In this study, we performed binary classification of construction machinery from satellite images. using transfer learning. In the field of artificial intelligence where the new and powerful models are published very often, authors of the study selected the most efficient models that are commonly used in transfer learning, i.e. AlexNet [8], VGG [10], ResNet [34,35], MobileNet [36,37], DenseNet [38], EfficientNet [39]. We evaluated all models in terms of various performance metrics. Additionally, we evaluated the best model in terms of both efficiency and accuracy using a case example. We analyzed the final model in a case study where detection of aggregate mining is discussed along the Thamirabarani River basin.

2. Materials and methods

In this study, we performed construction machinery image classification using satellite images. The dataset consists of different construction machinery images located on different ground types. It was created from scratch using Google Earth. A total of 3250 satellite images from various locations were split into training, validation, and test datasets as shown in Table 1. Properties of the images in the dataset are also given in Table 2. Images containing any type of construction machinery were grouped in the Construction Machinery (CM) category (positive group). Images collected for the dataset contain excavators, backhoes, dumpers, bulldozers, articulated haulers, graders, mobile cranes, wheel loaders, skid steer loaders, boom lifts for positive class. We collected images from different backgrounds for both categories. We collected images of cars and other non-construction vehicles, and other objects that might look like construction machinery for the No Construction Machinery (NCM) category (negative group). Consequently, we aimed the model to be able to distinguish the main features of construction machinery in any setting or background type. Additionally, we collected images in the dataset from different camera object distances between 30 m and 300 m. The images collected have approximately the same size but the target feature of the classification in the picture can occupy different percentages of the original image. Consequently, we wanted the model to classify a wider range of satellite images. We preprocessed the images by first resizing all dataset to 256 × 256, which is the standard for the ImageNet database. Additionally, for the training dataset, we applied random flip, random rotation, and color jitter for augmentation purposes.

We first selected six well-established architectures and then selected a specific version from these architecture groups after an extensive preliminary search. We utilized AlexNet, VGG16, ResNet101, MobileNetV2, DenseNet161, EfficientNetB0 architectures. Google's EfficientNets, are known to have both high accuracy and efficiency [39]. High accuracy values were reported by using very deep convolutional networks with VGGS or Visual Geometric Group [10]. AlexNet is still very popular despite being a relatively older model. It provides faster training mainly due to it uses ReLU as activation function [8]. DenseNets has high efficiency in terms of the number of parameters and depth. Compared to traditional models, DenseNet architecture needs fewer parameters for model training and uses the number of parameters more efficiently for a similar accuracy level. Each layer in the model has direct access from loss function to gradients. This flow of information makes DenseNets deeper and easier to train [38]. ResNet is one of the most popular models in the literature due to its high accuracy in image classification domain. Although ResNet has a deep architecture, it has lower complexity than other deep models and is easier to optimize. Therefore, deep models can be trained without increasing the training error [34,35]. MobileNets are designed for mobile and embedded vision applications, they have relatively less trainable parameters, so they train faster [36].

We trained the final six architectures using both transfer learning and random initialization. When we utilized transfer learning, the

Table 1
Dataset details.

Image classes	Training	Validation	Testing	Total
Construction Machine	1364	250	252	1866
No Construction Machine	1012	186	186	1384

Table 2

Properties of images in the dataset.

Image classes	Maximum height	Minimum height	Maximum width	Minimum width
Construction Machine	810	77	1846	69
No Construction Machine	807	60	1821	83

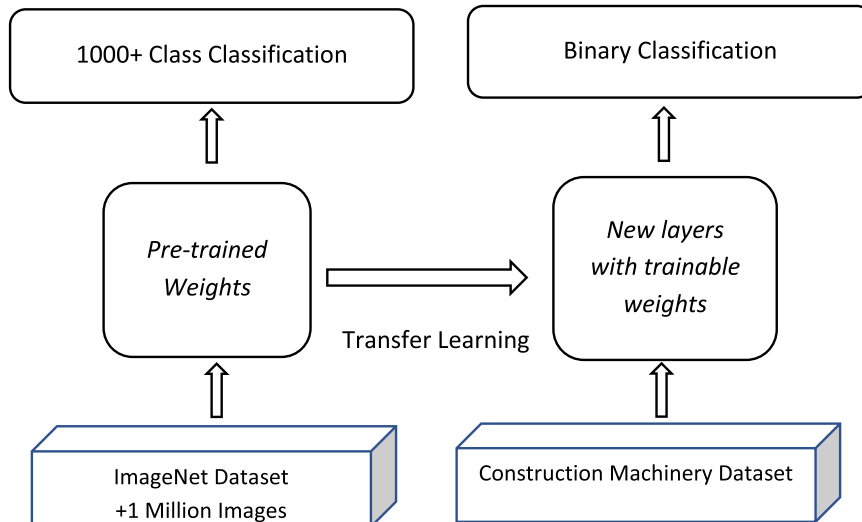
weights of the models transferred directly from ImageNet for the transfer learning method as seen in Fig. 1. We added final layers as fully connected dense layers for the classification task as seen in Fig. 2. When we utilized random initialization as the training scheme, the architecture of the model and the added final classification layer stayed the same as it was for the transfer learning method. The only difference was weights in the entire model were trainable and they were initialized randomly. Consequently, in the random initialization method, trainable parameters were much higher compared to transfer learning method in all architectures as seen in Table 3. We used the Pytorch library in python for models and their training. All models were trained with images from varying camera object distances. The setup used was Intel(R) Core (TM) i7-7700HQ CPU @ 2.80 GHz processor, 16.0 GB RAM, and NVIDIA GeForce GTX 1070 8 GB GPU for the training of all models.

We selected the main hyperparameters by utilizing a grid search. The main hyperparameters are shown in Table 4. We trained all models with two different optimizers and different batch sizes using different learning rates. We used SGD and Adam optimizers because they were the most commonly utilized optimizers in the image classification domain. Batch size of the models had to be limited because number of trainable parameters was much higher in random initialization models which led to capacity problems for the computer setup. Consequently, the highest batch size that can be trained without capacity surge was 16 and it was selected for the training of all models. SGD algorithm yielded better validation results for all models, so it was selected to be used for further analysis. As for the learning rate, the highest accuracy is obtained for 0.01 for all the models except AlexNet. We used a learning rate of 0.005 for the AlexNet model.

We utilized the most common metrics for evaluating model performances. Precision, recall, accuracy, and F-score are the most commonly used metrics in the image classification domain. Precision indicates percentage of true positives among all images that are classified as positive. Recall is percentage of true positives among all images that were actually positive. F-score is an evaluation parameter calculated with the harmonic mean of precision and recall [40]. It is a measure of a model's accuracy. Although F score is commonly used as F1 score, the general F score is F_β , where value of β can be used according to how much more important recall is than precision. As seen from Eq. (1) as the value of β increases above 1, the value of F_β signifies recall higher than precision.

$$F_\beta = (1 + \beta^2) \frac{\text{precision} \bullet \text{recall}}{\beta^2 \bullet \text{precision} + \text{recall}} \quad (1)$$

Considering the scope of the work accuracy, precision, recall, and training and test durations are used for comparison of models. Additionally, when detecting construction machinery from satellite images, especially for applications such as illegal mining, recall value being high will be relatively more critical than precision value. Consequently, we have also calculated F1 and F2 scores for all the model runs.

**Fig. 1.** Transfer learning method used for binary classification of images in our dataset.

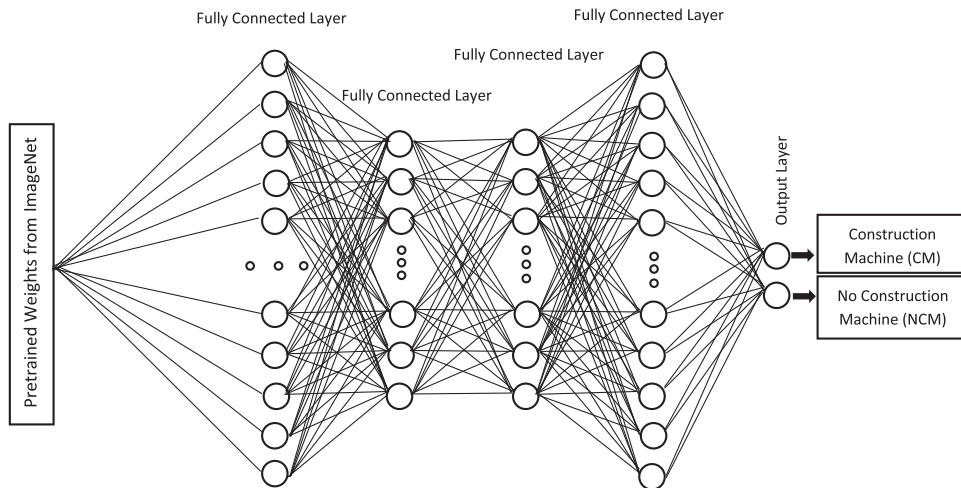


Fig. 2. Fully connected layers added for the classification task.

Table 3
Number of trainable parameters for each learning scheme.

Architecture	Learning scheme	Trainable parameters
EfficientNet	Transfer Learning	941,414
	Random Initialization	4,948,962
VGG	Transfer Learning	15,511,910
	Random Initialization	30,226,598
AlexNet	Transfer Learning	5,798,246
	Random Initialization	8,267,942
DenseNet	Transfer Learning	1,509,350
	Random Initialization	27,981,350
ResNet	Transfer Learning	1,411,430
	Random Initialization	43,911,590
MobileNet	Transfer Learning	941,414
	Random Initialization	3,165,286

Table 4
Hyperparameter search space during preliminary analysis.

Hyperparameter	Learning rate	Optimizer	Batch size
Search Space	Max: 0.05	SGD	Max: 64
	Min: 0.001	Adam	Min: 8

3. Results and discussion

All models are trained using the same train dataset, test accuracy of each model was determined using the test dataset. Table 5 shows average values for accuracy and loss in both training, validation, and test datasets obtained from all models. Duration of training

Table 5
Performance metrics -average of all models.

Average		Total duration (s)	Accuracy %	Loss
Training	Total	1028	89.72	0.25
	Transfer Learning	792	93.12	0.17
	Random Initialization	1264	86.32	0.33
Validation	Total	–	84.16	0.37
	Transfer Learning	–	88.61	0.27
	Random Initialization	–	79.72	0.47
Test	Total	6.87	87.42	0.38
	Transfer Learning	6.59	89.51	0.22
	Random Initialization	7.14	85.34	0.54

is longer for randomly initialized models compared to transfer learning as expected. It was also observed that the time required for classification of images is slightly less for models that used random initialization. Average accuracy of all models trained with transfer learning is higher compared to random initialization. This holds true for training, validation, and test phase. The difference between average losses of these two learning methods is even more prominent, especially for test phase. Average test accuracy is around 87%.

Training durations per epoch and time elapsed for the evaluation of test data are given in Fig. 3. AlexNet trained with transfer learning (TL) has the shortest training duration per epoch closely followed by MobileNet and EfficientNet. The same models also evaluate the test data fastest. DenseNet run the least efficiently for both training and evaluation. When the accuracies of the trained models are compared DenseNet161, ResNet101 and VGG16 have the highest test accuracies that are above 0.9 for transfer learning models (Fig. 4). DenseNet161 also has the highest test accuracy for Randomly initialized models, only one that is above 0.85. Training accuracies are slightly higher than test accuracies for most of the models as would be expected.

Both F1 and F2 scores of all models are compared in Fig. 5. F1 score indicates model performance attaining equal importance to recall and precision. When the models are utilized in detecting construction activities such as illegal aggregate mining false negatives will be more problematic compared to false positives. When a different type of machinery is detected as a positive, a quick scan with the human eye can easily rule the prediction out. However, if the model classifies an image containing construction machinery as in the negative class that would mean failing the job it was built for. For that reason, F2 scores are also calculated. DenseNet161 yields the highest F1 and F2 scores for transfer learning. DenseNet's F1 score in transfer learning model is the only core that is higher than 0.9. ResNet101 and VGG16 closely follow DenseNet161 for F2 and F1 scores respectively. Transfer learning outperforms random initialization models for all models and both F scores.

Table 6 shows two images, one containing construction machinery, the other having very similar features but does not contain construction machinery. We selected both images to have features that are relatively hard to correctly classify. As can be seen from the table, the first image is only correctly classified only by DenseNet161 and ResNet101. The second image is correctly classified by three out of six models, i.e. DenseNet161, VGG16, and MobileNetV2. For a construction machinery detection task such as monitoring illegal aggregate mining, correct classification of positives (CM) is more critical compared to the correct classification of negatives (NCM). Therefore, DenseNet161 and ResNet101 architectures are better suited for such a task.

Even though most model performances are close to each other, DenseNet161 trained with transfer learning also returns the highest precision and recall values; 0.892 and 0.953 for test data respectively. DenseNet161 is closely followed by VGG16 with the second highest precision; 0.889, however, the second highest recall belongs to ResNet101 with 0.943. In the light of the above discussions ResNet101 and DenseNet161 models are selected for further evaluation. Confusion matrices for these two models are given in Fig. 6. DenseNet161 has slightly better evaluation metrics, there are only 9 instances where the model labeled an image containing construction machinery as negative out of 438 test samples similar to ResNet101. On the other hand, DenseNet161 labeled 26 NCM images as construction machinery, this number is 33 for ResNet101 faring worse than DenseNet architecture.

4. Case study

In order to analyze efficiency of models for a real-life application we discussed a case study where an area known to have aggregate mining activities needs to be scanned. We determined additional parameters for the models to be able to use them in calculations. Initially we wanted to determine the relationship between the accuracy of the model and the area covered in one image. Accordingly, we performed calculations in our case study after optimizing this relationship.

The dataset that was created for training, validation, and testing of data contain images collected from google earth between 30 m and 300 m camera object distance. Images in this dataset have similar number of pixels however number of pixels of a construction machinery is different for different camera object distances. When satellite imagery is purchased with a specific resolution, it needs to be subdivided into multiple images before being classified by our models. Each image should contain a minimum number of pixels occupied by construction machinery because as the number of pixels occupied by the object reduces accuracy also reduces [41]. We decided to further evaluate these two models to see how much accuracy will reduce when the machinery occupies fewer pixels in the

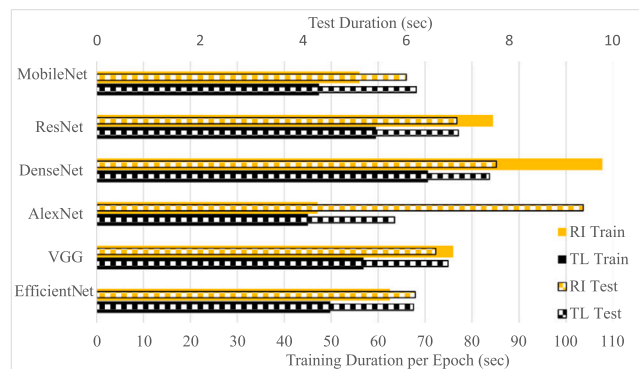


Fig. 3. Time required per epoch for training and evaluation duration with test data for models trained with Random Initialization (RI) and Transfer learning (TL).

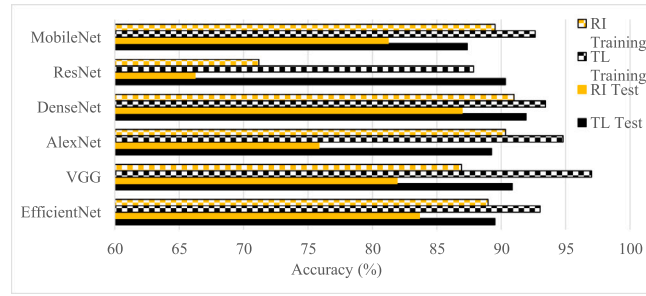


Fig. 4. Training and test accuracies for models trained with Random Initialization (RI) and Transfer learning (TL).

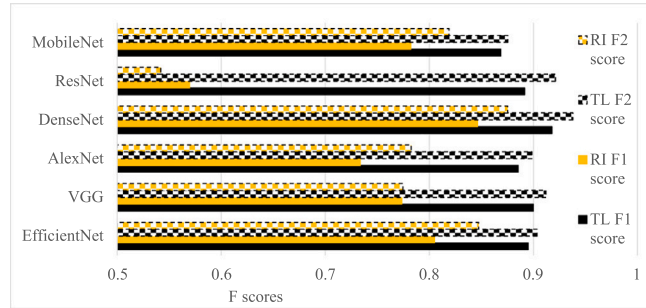


Fig. 5. F1 and F2 scores for models trained with Random Initialization (RI) and Transfer learning (TL).

image.

Consequently, we performed a second test run with DenseNet161 and ResNet101 using images taken from different camera object distances. Five new test datasets are created each containing 60 images from Google Earth. Datasets contain images from camera object distances 100, 200, 300, 400, and 500 m. Each new dataset contains 30 images with construction machinery and 30 more without construction machinery. When new datasets are fed to these models, accuracy and precision reduce with camera object distance as expected as seen in Table 7. Both 400 m and 500 m datasets contain images that construction machinery pixels are less than the minimum of the training dataset. Consequently, a decrease in accuracy is observed to be sharper after 300 m dataset. DenseNet161 model's accuracy stays above 90% even in 500 m dataset.

DenseNet shows promising accuracy and precision with scores around 0.95 at this camera object distance. Recall value of ResNet101 in 500 m dataset is higher than DenseNet161, whereas it is vice versa for precision values. There is a large difference between F1 scores but F2 scores are very similar for the same 500 m dataset. All performance metrics for 300 m dataset are quite close to each other. Reduction of these performance scores is emphasized in 400 m dataset, as expected. The most notable difference is precision of ResNet101 reduces drastically in 500 m dataset. Considering these datasets have 60 images each and the original test dataset has 438 images, new datasets are less representative compared to the initial test dataset. However, it can still be concluded that DenseNet161 performs better even when the number of pixels occupied by construction machinery is reduced below the minimum of the training dataset.

As explained in the introduction chapter this paper also aims to explore the applicability of the construction machine detector for monitoring construction related activities for large territories. All models are trained and tested using a dataset with images from 30 m up to 300 m camera object distances in google earth view. A representative calculation is performed to illustrate how DenseNet161 model can be utilized. We assumed pixels occupied by construction machinery in the example calculation images should be equivalent to that of images obtained from 300 m camera object distance in google earth for optimum efficiency. The 128-km-length Thamirabarani River in the Tamil Nadu, state of India was chosen as the target region. Thamirabarani River basin is known for intense illegal aggregate mining activities [42]. Illegally mined aggregates are used for concrete production throughout the region.

Using satellite images from 300 m camera object distance approximate area covered in one picture is around 0.2 km^2 . Entire Basin of Thamirabarani River is reported as 4400 km^2 . Using this information, we calculated that our trained DenseNet161 model can scan the entire region with approximately 22,000 images. The time required to scan the entire Thamirabarani River for construction machinery detection is around 374 s with the current setup. The accuracy of DenseNet161 for 300 m camera object distance dataset is 0.967. Accuracy of DenseNet model on the original test set is 0.919, which contains more images but images with varying camera object distances. This would mean using the proposed model for fast and accurate detection of construction machinery could offer a low-cost solution for the detection of illegal concrete aggregate mining activities.

Table 6

Classification performance of transfer learning models on two example images.

						
An image containing construction machinery-class positive (CM)						
Architecture:	EfficientNet	VGG	AlexNet	DenseNet	ResNet	MobileNet
Prediction:	NCM	NCM	NCM	CM	CM	NCM
						
An image that does not contain construction machinery-class negative (NCM)						
Architecture:	EfficientNet	VGG	AlexNet	DenseNet	ResNet	MobileNet
Prediction:	CM	NCM	CM	NCM	CM	NCM

5. Conclusion

The efficiency of construction activity monitoring is dependent on the size of the construction field. For small to medium-sized construction sites, construction activities can be monitored either through manpower or with a hybrid system of drones and manpower. When larger areas need to be covered, drone operations can become ineffective due to their short ranges. Additionally, time consumption and cost increase sharply with a small reduction in monitoring efficiency for large size construction sites. Consequently, using satellite images can be a good alternative for construction tasks spanning large areas such as aggregate mining operations in infrastructural construction. Such a monitoring task can easily be performed through image classification algorithms by

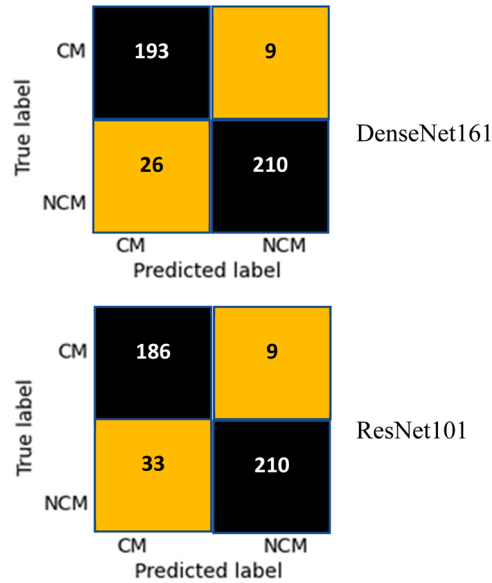


Fig. 6. Confusion matrices of DenseNet161 and ResNet101 for test data.

Table 7

Performance metrics vs camera object distance in the new datasets.

Architecture	Dataset (m)	Accuracy	Loss	Precision	Recall	F1 score	F2 score
DenseNet	100	0.967	0.109	1.000	0.929	0.961	0.941
	200	0.967	0.076	0.967	0.967	0.967	0.967
	300	0.967	0.147	0.933	1.000	0.962	0.983
	400	0.933	0.177	0.902	0.971	0.931	0.954
	500	0.900	0.308	0.848	0.956	0.889	0.925
ResNet	100	0.967	0.109	1.000	0.935	0.986	0.947
	200	0.933	0.136	0.967	0.923	0.954	0.929
	300	0.967	0.194	0.948	1.000	0.958	0.989
	400	0.917	0.277	0.889	0.962	0.900	0.943
	500	0.850	0.325	0.774	0.973	0.804	0.921

training the algorithms to detect construction machinery.

In this study, we performed construction machinery detection with satellite images. We created a dataset from scratch using Google Earth satellite images that contains construction machinery to train convolutional neural networks. Six different architectures were trained using both randomly initialized weights and transfer learning to classify satellite images. A binary classification task is used to detect whether or not the images contained construction machinery. We created a dataset consisting of 3250 satellite images from Google Earth with camera object distances between 30 m and 300 m. We identified DenseNet161 and ResNet101 for their higher precision and lower recall values among the initial six architectures.

We applied our models in a case study where we analyzed efficiency of the best performing model in a scenario where we wanted to detect aggregate mining used for concrete production in a river basin. Consequently, we first analyzed the two top performing models, using new datasets created with images with 100, 200, 300, 400, and 500 m camera object distances. Model performances reduce when the object, i.e. construction machinery, occupied a smaller area in the original image. DenseNet161 provided the highest accuracy and precision for all test datasets. We found that 300 m is an optimum distance where an area of 0.2 km² is covered in one image and accuracy is still high. The case study concluded using DenseNet161 an entire river basin can be scanned within 374 s with high accuracy and low recall. This classification rate might also enable real-time classification which can lead to efficient surveillance of large regions with a slightly better computational set up. Detecting construction activities with satellite imageries in a low-cost manner without the need for manpower has the potential to help increase the sustainability of construction activities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

Data on the Google Earth images openly available at <https://earth.google.com/web/> further inquiries can be directed to the corresponding author.

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